

Chapter 03: Higher-Order Differential Equations & Oscillators

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Applied Ordinary Differential Equations (ENGR 213) | Textbook: Dennis G. Zill (7th Ed., Chapter 3)

1. Executive Summary & First-Principles Philosophy

Higher-order differential equations govern all vibrating engineering structures: car suspensions, skyscraper wind dampening, airplane wing flutter, and RLC electric circuits. The centerpiece of Chapter 3 is the **Linear Superposition Principle**: the general solution to a non-homogeneous linear equation is:

$$y(x) = y_c(x) + y_p(x)$$

where $y_c(x)$ (the complementary function) solves the homogeneous equation $L[y] = 0$, and $y_p(x)$ is any particular solution solving $L[y] = g(x)$.

2. Core Mechanics & Mathematical Engine

A. The Wronskian & Linear Independence

Two solutions $y_1(x)$ and $y_2(x)$ to a second-order homogeneous linear ODE are **linearly independent** on an interval I if and only if their Wronskian determinant is non-zero everywhere on I :

$$W(y_1, y_2)(x) = \begin{vmatrix} y_1 & y_2 & y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2 \neq 0$$

B. Constant-Coefficient Homogeneous Equations ($ay'' + by' + cy = 0$)

Substitute trial solution $y = e^{rx}$, giving the **characteristic auxiliary equation**:

$$ar^2 + br + c = 0$$

Three distinct physical cases arise from the roots $r_1, r_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

1. **Case 1: Distinct Real Roots** ($b^2 - 4ac > 0$):

$$y_c(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

2. **Case 2: Repeated Real Root** ($b^2 - 4ac = 0, r_1 = r_2 = r$):

$$y_c(x) = c_1 e^{rx} + c_2 x e^{rx}$$

3. **Case 3: Complex Conjugate Roots** ($b^2 - 4ac < 0, r = \alpha \pm i\beta$): Using Euler's formula $e^{(\alpha \pm i\beta)x} = e^{\alpha x}(\cos(\beta x) \pm i \sin(\beta x))$:

$$y_c(x) = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

C. Method of Undetermined Coefficients

Used when the driving term $g(x)$ is a polynomial, exponential, sine/cosine, or linear combinations/products thereof:

- **The Duplication / Modification Rule:** If any term in the trial particular solution $Y_p(x)$ duplicates a term in the complementary solution $y_c(x)$, that term (and its associated block) must be multiplied by x^k , where k is the smallest positive integer that eliminates the duplication.

D. Method of Variation of Parameters (Universal Method)

Works for **any** continuous driving function $g(x)$ (including $\tan(x)$, $\sec(x)$, $1/x$): Given standard form $y'' + P(x)y' + Q(x)y = f(x)$:

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$$

where:

$$u_1'(x) = -\frac{y_2(x)f(x)}{W(y_1, y_2)}, \quad u_2'(x) = \frac{y_1(x)f(x)}{W(y_1, y_2)}$$

3. Curriculum-Grounded Visual Reference

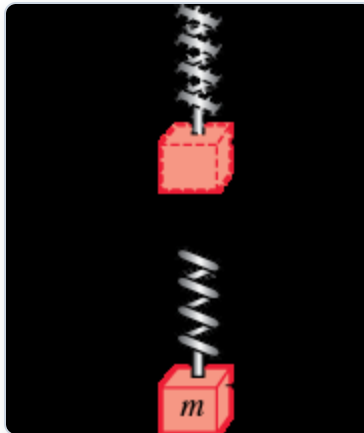


Figure 3.1: Mass-spring equilibrium displacement coordinate system — from Zill 7th Ed. Chapter 3 (Fig. 3.8.2).

$$x(t) = \frac{\sqrt{3}}{6} \cos(8t - (-0.245)) \quad \text{or} \quad x(t) = \frac{\sqrt{3}}{6} \cos(8t + 0.245). \quad (9') \equiv$$

■ Graphical Interpretation

FIGURE 3.8.4(a) illustrates the mass in Example 2 going through approximately two complete cycles of motion. Reading left to right, the first five positions marked with black dots in the figure correspond, respectively,

- to the initial position ($x = \frac{2}{3}$) of the mass below the equilibrium position,
- to the mass passing through the equilibrium position ($x = 0$) for the first time heading upward,

Figure 3.2: Damped oscillator decay curves (Underdamped vs Critically Damped vs Overdamped) — from Zill 7th Ed. Chapter 3 (Fig. 3.8.4).

4. Mechanical Oscillations: Classification of Damping

For a mass-spring-damper system governed by $mx'' + \beta x' + kx = 0$:

$$\omega_0 = \sqrt{\frac{k}{m}} \quad (\text{Natural Frequency}), \quad 2\lambda = \frac{\beta}{m}$$

1. **Overdamped** ($\beta^2 - 4mk > 0$): Two distinct negative real roots. The system slowly returns to equilibrium without oscillating.
2. **Critically Damped** ($\beta^2 - 4mk = 0$): Repeated real root. Returns to equilibrium in the shortest possible time without oscillating.
3. **Underdamped** ($\beta^2 - 4mk < 0$): Complex roots $\lambda \pm i\omega_d$. Oscillates with decaying amplitude envelope $x(t) = Ae^{-\lambda t} \cos(\omega_d t - \phi)$, where quasi-frequency $\omega_d = \sqrt{\omega_0^2 - \lambda^2}$.

5. Rapid Exam Traps & Red Flags

- **⚠ Trap 1: Leading Coefficient in Variation of Parameters:** Standard form requires leading coefficient 1: $y'' + Py' + Qy = f(x)$. If the problem starts as $a(x)y'' + \dots = g(x)$, you **must divide through by** $a(x)$ so that $f(x) = g(x)/a(x)$.
- **⚠ Trap 2: Forgetting the Duplication Rule:** If $y_c = c_1 e^{2x} + c_2 x e^{2x}$ and $g(x) = e^{2x}$, guessing $Y_p = Ae^{2x}$ will completely collapse ($0 = e^{2x}$). You must multiply by x^2 : $Y_p = Ax^2 e^{2x}$.